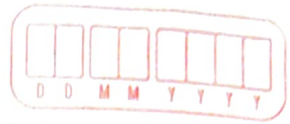
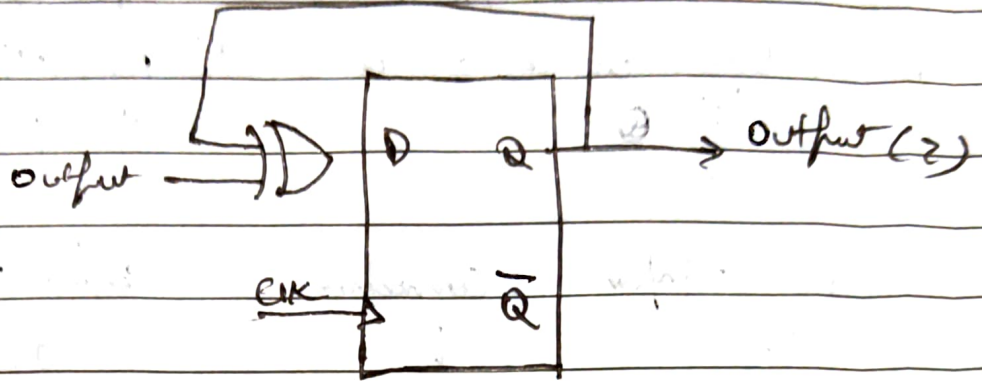


Assignment - Day 4



Q 1) Explain the behaviour of the given circuit



XOR gate inputs are

External input = x

Feedback input = Q

$$D = x \oplus Q$$

$$Q_{next} = D$$

$$Q_{next} = x \oplus Q$$

$$Z_{next} = x \oplus Z$$

State table

External input = x	Present state (Q) = (x)	Q
Feedback input = Q	0	0
	0	1
	1	1
	1	0

Case 1: $x = 0 \Rightarrow Q_{next} = Q$

Case 2: $x = 1 \Rightarrow Q_{next} = \bar{Q}$

The circuit behaves like a T flip-flop.

Q)2 Design a posedge triggered DFF using 2:1 Muxc
 only master latch $\Rightarrow m = \overline{clk} D + clk M$

by slave latch $\Rightarrow Q = clk M + \overline{clk} Q$

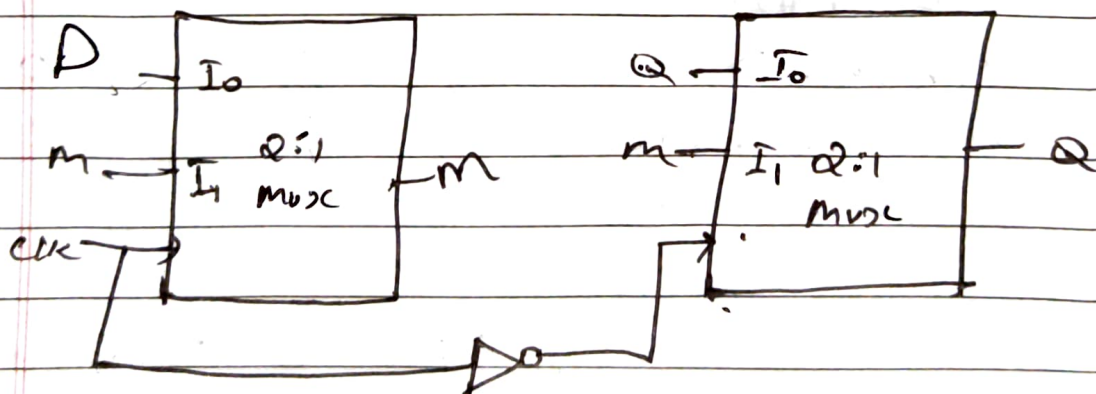
$clk = 0$

↓

$clk = 1$

↓

mux input	connection	mux output	connection
I_0	D	I_0	Q
I_1	M (feedback)	I_1	m
select	clk	select	clk
output	m	output	Q



Q) 3. Design a negedge Triggered T.FF using 2:1 mux
 only characteristic equation of TFF is $Q_{next} = T \oplus Q$

for 2:1 mux
 select line = T

$$I_0 = Q$$

$$I_1 = \bar{Q}$$

$$y = \bar{T}Q + T\bar{Q}$$

$$y = T \oplus Q$$

$$M = CLK M + CLK (T \oplus M)$$

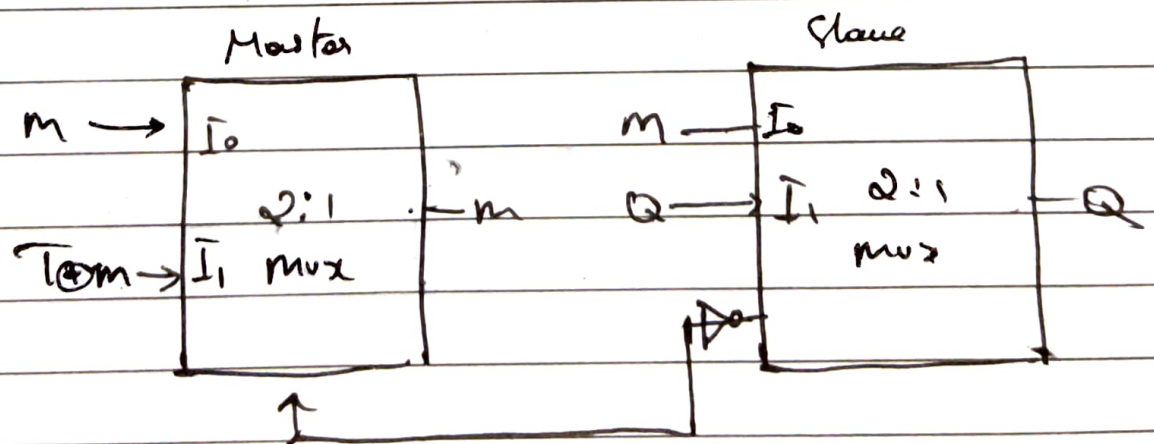
$$Q = CLK Q + CLK (\bar{M})$$

Master latch CLK = 1

Slave latch CLK = 0

I/k	connection
I_0	M (feedback)
I_1	$T \oplus M$
Select	CLK
output	M

I/k	connection
I_0	M
I_1	Q (feedback)
Select	CLK
output	Q



Q) 4 Convert by adding external gates

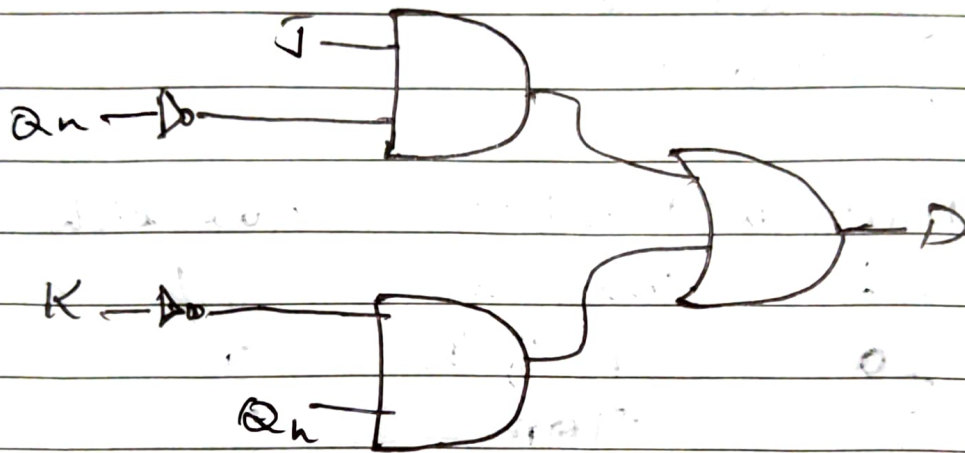
a) A DFF to JK-FF

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

Ans:-

$$Q_{n+1} = D$$

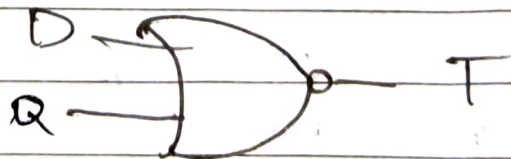
$$D = J\bar{Q}_n + \bar{K}Q_n$$



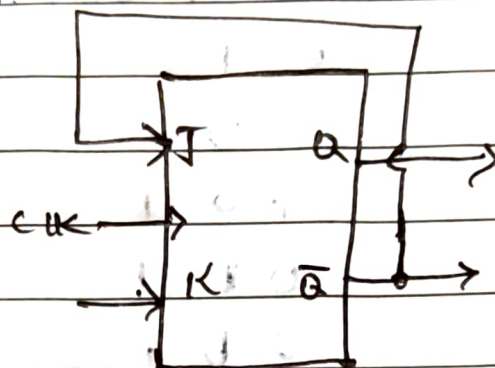
b) A T-FF to a D-FF

$$Q_{n+1} = T \oplus Q \quad \cdot \quad Q_{n+1} = D \quad \Rightarrow \quad D = T \oplus Q$$

$$T = D \oplus Q$$



Q) 5 In a positive edge triggered JK FF, we have $J = \bar{Q}$ & $K = 1$ as shown in the fig. Assume the flip flop was initially cleared and then clocked for 6 clk pulses. Find the sequence at the output of these 6 clock cycles.



Ans $Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$
 $K = 1 \Rightarrow \bar{K} = 0 \quad J = \bar{Q}$

$$Q_{n+1} = \bar{Q} \cdot \bar{Q} + 0$$

$$Q_{n+1} = \bar{Q} \quad (\text{Toggles on every clock pulses})$$

clk pulse	present Q	$J = \bar{Q}$	K	Next Q
0	0	1	1	1
1	1	0	1	0
2	0	1	1	1
3	1	0	1	0
4	0	1	1	1
5	1	0	1	0
6	0	1	1	1

Sequence is $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0$